



## Subsets, Supersets & Power Sets

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SUBJECT

Mathematics Foundation T

SEMESTER

Semester 1

### // Notes Summary

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{
  title: "Subsets, Supersets Power Sets" semester:
  subject: "Mathematics Foundation to Computer Science
  I" subjectSlug: "mathfoundation1" unit: "Unit I:
  Set, Relation and Function" unitSlug:
  "unit1setrelationfunction" order: description:
  "Subset inclusion, proper subsets, supersets, set
```

# NOTES THAT STICK.

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Topics Covered

Intermediate

Difficulty Level



## Why this topic exists

Once sets are defined, the primary focus shifts to comparing them and modeling their relationships. Subset relationships determine whether one set is contained within another, identical, or structurally equivalent. In computer science, these concepts map directly to database queries (where a filtered view or table query represents a subset of a full dataset), type systems (subtyping in OOP), and state space analysis.

## Subset ( $A \subseteq B$ )

### ✦ [DEFINITION] Formal Subset Condition

Set  $A$  is a subset of set  $B$  if every element of  $A$  is also an element of  $B$ .

$$A \subseteq B \iff (\forall x)(x \in A \implies x \in B)$$

**Example:** If  $A = \{1, 2\}$  and  $B = \{1, 2, 3, 4\}$ , then  $A \subseteq B$ .

### ✓ [VALID / RULE] Fundamental Subset Properties

1. **Reflexivity:** Every set is a subset of itself:  $A \subseteq A$ . 2. **Vacuous Inclusion:** The empty set is a subset of every set:  $\emptyset \subseteq A$ .

## Proper Subset ( $A \subset B$ )

### ✦ [DEFINITION] Proper Subset

Set  $A$  is a subset of  $B$ , but  $A$  is strictly not equal to  $B$ . Set  $B$  contains at least one element that is not in  $A$ .

$$A \subset B \iff (A \subseteq B \wedge A \neq B)$$

- **Example:** If  $A = \{1, 2\}$  and  $B = \{1, 2, 3\}$ , then  $A \subset B$ . - **Distinction:**  $A \subseteq B$  permits  $A = B$ , whereas  $A \subset B$  strictly prohibits equality.



## Superset ( $A \supseteq B$ ) & Proper Superset ( $A \supset B$ )

- **Superset ( $A \supseteq B$ ):** Set  $A$  contains every element of set  $B$ .
- **Proper Superset ( $A \supset B$ ):** Set  $A$  contains every element of  $B$  and has at least one additional element.

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## Equal Sets vs. Equivalent Sets

#### Equal Sets ( $A = B$ ) - **Definition:** Two sets are equal if they contain the exact same elements regardless of order or listing frequency. - **Formal Proof Standard:**  $A = B \iff (A \subseteq B \wedge B \subseteq A)$  - **Example:**  $A = \{1, 2, 3\}$  and  $B = \{3, 2, 1\} \implies A = B$ .

#### Equivalent Sets ( $A \sim B$ ) - **Definition:** Two sets are equivalent if they have the exact same cardinality (number of elements), regardless of their individual elements. - **Formal Notation:**  $A \sim B \iff |A| = |B|$  - **Example:**  $A = \{1, 2, 3\}$  and  $B = \{a, b, c\} \implies A \sim B$  (since  $|A| = |B| = 3$ ), though  $A \neq B$ .

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## Universal Set ( $U$ )

- **Definition:** The set containing all possible elements under consideration for a specific domain or context. Every set within that domain is a subset of  $U$ .
- **Example:** In an integer modular arithmetic problem,  $U = \mathbb{Z}$  (the set of all integers).
- **Significance:** The complement operation ( $A'$ ) is undefined without an explicitly bounded Universal Set  $U$ .

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## Power Set ( $\mathcal{P}(A)$ or $2^A$ )

### ♦ [DEFINITION] Power Set Definition

The set of all possible subsets of a given set  $A$ , including the empty set  $\emptyset$  and  $A$  itself.

$$\mathcal{P}(A) = \{S \mid S \subseteq A\}$$



**Example:** Given  $A = \{1, 2\}$ :

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

#### Cardinality Formula

If a set  $A$  has  $n$  elements ( $|A| = n$ ), its power set contains  $2^n$  elements:

$$|\mathcal{P}(A)| = 2^n$$

- **Combinatorial Reason:** For every element  $x \in A$ , there are exactly 2 choices when constructing a subset: either include  $x$  or exclude  $x$ . Across  $n$  independent elements:

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